

RATES Whitepaper

Stable Company

Version 0.1

June 2026

Abstract

RATES is the protocol token of USDx, with three core utilities: validators stake RATES to attest loan data and gate the minting of new USDx collateral; RATES stakers provide the first-loss safety module backstopping the reserve; and RATES holders govern a narrow set of protocol parameters. This paper specifies its distribution, supply schedule, emissions, validator mechanics, first-loss safety module, and governance in full. USDx itself is specified in the USDx whitepaper; this document assumes that context.

Detailed operational parameters — validator selection and slashing, delegation mechanics, governance procedures, validator onboarding, treasury management, and cross-chain accounting — are deferred to a follow-up implementation specification published ahead of RATES distribution.

Keywords: protocol token, validator staking, first-loss safety module, emissions schedule, governance, mortgage-backed stablecoin.

Table of Contents

1. Overview
2. Validator Staking
3. First-Loss Safety Module
4. Governance
5. Supply, Distribution, and Emissions
6. Topics Deferred to Follow-Up Implementation Specification

1. Overview

RATES has three core utilities: validators stake RATES to attest loan data and gate the minting of new USDX collateral; RATES stakers provide the first-loss safety module backstopping the reserve; and RATES holders govern a narrow set of protocol parameters. This paper specifies distribution, supply schedule, emissions, validator mechanics, and governance in full.

2. Validator Staking

Validators stake RATES to join the active validator set and are responsible for two classes of attestation that gate new USDX collateral: origination attestations (at the origination of a new tokenized mortgage, whether through Stable's channel or a partner pipeline) and quarterly attestations (re-attesting property valuation and lien status).

Scope. Validators attest to the cryptographic integrity of loan data produced by approved third-party providers — AVM vendors, title companies, credit bureaus, flood and tax data providers — rather than independently producing it. The role is verification-of-verification.

Economic structure. Validators earn modest flat or bps-capped fees per origination and per quarterly re-attestation, paid in USDX, priced below the combined cost of the traditional services they replace. 20% of validator USDX fees route to the RATES buyback-and-burn contract; the balance flows to the validator. Primary compensation is RATES emissions (Section 5), supplemented by safety module rewards.

Set composition. Initial active set of 21 validators at origination launch, scaling toward a steady-state cap. Minimum self-stake of 1,000,000 RATES per seat. Delegation supported.

Deferred to a follow-up implementation specification. Detailed validator selection criteria, delegation mechanics (commissions, minimums, unbonding), and the precise slashing schedule (offenses, magnitudes, dispute process) are specified in a follow-up implementation document published ahead of validator launch. These parameters are operationally critical but separable from the framework specified in this paper.

3. First-Loss Safety Module

A portion of staked RATES sits in the safety module — Tranche 1 of the USDX loss waterfall. On a reserve shortfall, safety-module stake is slashed pro rata to make USDX holders whole before any other tranche is affected. In exchange for first-loss exposure, stakers earn a share of protocol fees plus emissions.

Safety-module staking and validator staking are distinct activities backed by distinct stake pools. Validator self-stake lives in the validator contract and is exposed only to attestation-fault slashing; safety-module stake lives in the safety-module contract and is exposed only to loss-waterfall slashing. The two pools are independent: a validator's self-stake is not at first-loss risk in the waterfall, and safety-module stake is not at risk for attestation faults. A participant may operate a validator, stake into the safety module, or both — each choice is opted into separately and slashed only from the pool to which it belongs.

4. Governance

RATES holders govern a narrow set of parameters: approved AVM vendors, approved repo counterparties, loan product whitelisting, validator set size, and slashing parameters. Vote weight is stake-weighted with a lock-duration multiplier (ve-style). Governance scope is intentionally limited; operational decisions remain with Stable Company as the licensed operator.

Emergency response. Pausing mints, pausing redemptions, halting the autonomous peg-maintenance bot (APMB; USDX whitepaper §7.4), and triggering a loss-waterfall draw are reserved to Stable Company, publicly logged and subject to retrospective review.

Deferred to a follow-up implementation specification. Governance procedures — proposal thresholds, voting periods, quorum, timelock specifics, and the precise boundary between RATES governance and operator authority — are specified in a follow-up implementation document published ahead of token distribution.

5. Supply, Distribution, and Emissions

This section is the canonical specification of RATES supply. It specifies initial supply, distribution, the emissions schedule, and supply dynamics. Additional supply is created post-genesis only via the loss waterfall (USDx whitepaper Section 8) and the emissions schedule below. Tokens burnt via the buyback mechanism (Section 5.4) reduce circulating and total supply; scheduled emissions withheld by the activity-gating mechanism (Section 5.5) are forgone — never minted — and so reduce realized supply growth relative to the schedule.

5.1 Initial Supply

The total initial supply of RATES at genesis is 1,000,000,000 tokens (1B). The full allocation across protocol, operator, investor, team, treasury, safety-module seed, and community categories — together with vesting schedules — is specified in the implementation specification published ahead of RATES distribution. A portion of initial supply is staked in the Tranche 1 safety module at genesis to provide immediate protocol-owned backstop capacity; this stake is slashed pro rata alongside external safety-module stakers in any loss event.

5.2 Emissions Schedule

Beyond the initial 1B supply, RATES are emitted on a geometrically decaying annual schedule applied to the initial 1B supply. The year-one emission rate is 12% of the initial supply. Each subsequent year's rate is 88% of the prior year's rate. Formally, the year- n emission rate is $r_n = 0.12 \cdot 0.88^{n-1}$, and the year- n emission amount (in tokens) is $E_n = r_n \cdot S_0$, where S_0 is the genesis supply of 1B RATES. The emission base does not compound: each year's emission is a fixed share of the original 1B, not of the prevailing outstanding supply.

Year	Rate	Emission (M)	End-of-year supply (M)
1	12.00%	120.0	1,120.0
2	10.56%	105.6	1,225.6
3	9.29%	92.9	1,318.5
5	7.20%	72.0	1,472.3
10	3.80%	38.0	1,721.5

20	1.06%	10.6	1,922.4
----	-------	------	---------

The sum of future emission rates over infinite time converges to exactly 100% of initial supply: $\sum_{n=1}^{\infty} 0.12 \cdot 0.88^{n-1} = 0.12 / (1 - 0.88) = 1$. Cumulative emissions therefore total exactly 1B RATES, capping total supply at exactly **2B RATES** under the emissions schedule alone, absent waterfall mints and buyback burns.

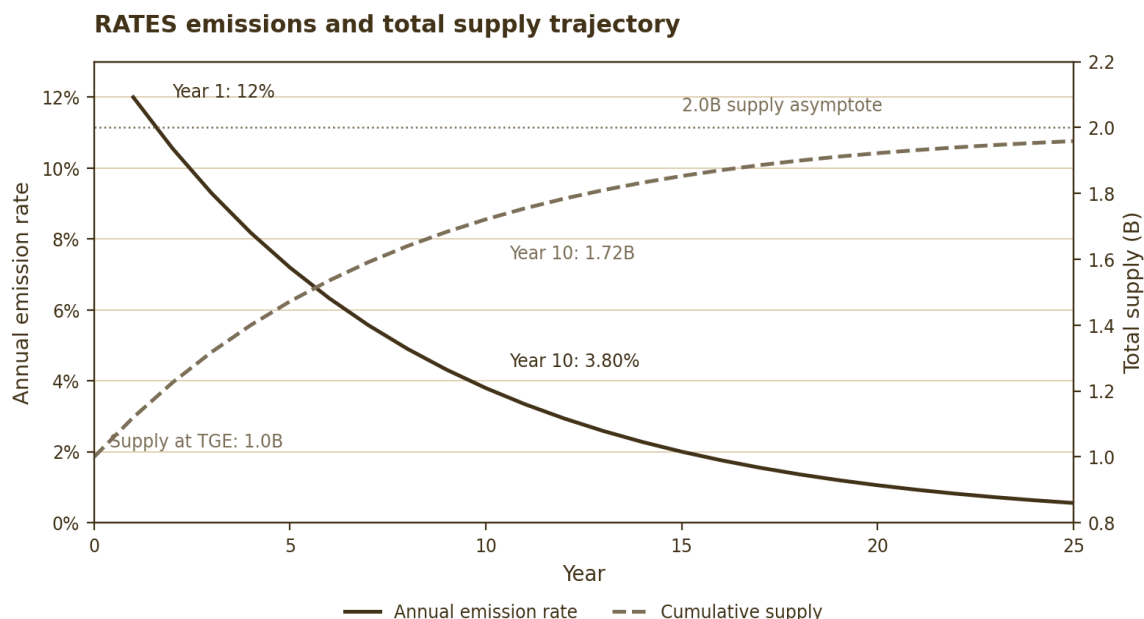


Figure 1: Annual RATES emission rate (decaying at 0.88 ratio) and cumulative supply (asymptoting at 2B) over the first 25 years.

5.3 Emissions Split

Emissions are split between validators and tranche stakers as follows:

- **50% to validators** — distributed pro rata based on attestations signed during the emission epoch. Validators receive this emission in addition to the per-origination and per-attestation USDX fees (Section 2).
- **50% to tranche stakers** — sub-allocated by risk seniority:
 - 60% (30% of total emissions) to Tranche 1 (RATES safety module) stakers
 - 25% (12.5% of total emissions) to Tranche 2 (stablecoin vault) stakers
 - 15% (7.5% of total emissions) to Tranche 3 (mUSDX) stakers

The split parameters are adjustable by governance within protocol-defined bounds.

5.4 Buyback and Burn

20% of validator USDX fees are automatically allocated to the buyback-and-burn contract, which uses these proceeds to purchase RATES on the open market and burn the acquired tokens. Validators earn 80% of their stated USDX fees on attestations; the remaining 20% is deducted pre-distribution and routed to the buyback contract. The effect is to tie RATES supply dynamics directly to protocol economic activity: every origination attestation and every quarterly re-attestation generates buyback demand. The mechanism is analogous to the base-fee burn in Ethereum (EIP-1559) — usage creates deflationary pressure on the token.

5.5 Activity-Gated Emissions

Validator emissions are gated by validator activity — specifically, the number of unique properties attested in a given month. An attestation event is either an origination attestation (new mortgage admitted to the reserve) or a quarterly re-attestation (property value and lien status refreshed). The gating mechanism operates on a linear scale between a minimum and maximum threshold:

- **Below the minimum threshold:** 0% of scheduled validator emissions are distributed. The full scheduled amount is forgone — never minted.
- **At the minimum threshold:** 10% of scheduled validator emissions are distributed to active validators pro rata.
- **Between minimum and maximum:** the distributed fraction grows linearly from 10% at the minimum to 100% at the maximum. Any undistributed fraction is forgone — never minted.
- **At or above the maximum threshold:** 100% of scheduled emissions are distributed.

Year-1 thresholds: minimum of 100 attested properties per month, maximum of 1,000.

Threshold growth: the minimum and maximum grow by 10% annually until the minimum reaches 1,000 properties per month (approximately year 26 at 10% compounding). Once the minimum reaches 1,000, both thresholds lock at 1,000 and

10,000 respectively, maintaining the 10x ratio in steady state.

5.6 Tranche Coverage Bands

Each tranche operates within a coverage band defined by a minimum and a maximum, both expressed as percentages of outstanding USDX supply. The band governs how tranche emissions are split between stakers and capacity building (Section 5.7).

Tranche	Minimum coverage	Maximum coverage
Tranche 1 — RATES safety module	0.5%	10%
Tranche 2 — Stablecoin vault	1%	10%
Tranche 3 — mUSDx	2%	10%

Coverage $c(T)$ for tranche T is computed as the USDx-equivalent value of capital staked in the tranche divided by the outstanding USDx supply. There is no cap on tranche coverage above the maximum; stakers may provide additional capacity freely, and the yield per staker naturally dilutes as the pool grows.

5.7 Emissions Behavior by Coverage

A tranche's allocated emissions are split between two destinations: staker rewards (distributed to tranche stakers pro rata) and capacity building (used to increase the tranche's protocol-owned capacity). The split depends on where current coverage sits within the tranche's band.

- **Below minimum coverage:** the tranche earns no staker rewards. 100% of its allocated emissions go to building capacity.
- **Between minimum and maximum:** the split scales linearly — 10% to stakers / 90% to building at minimum coverage, 100% to stakers / 0% to building at maximum coverage, linearly in between.
- **At or above maximum coverage:** 100% of allocated emissions go to stakers. No further capacity building occurs.

5.8 Permanent Protocol Capital

Capital acquired through the capacity-building mechanism is protocol-owned and permanent. It cannot be withdrawn by external parties and is moved only by governance action. Protocol-owned tranche capital is slashed pro rata alongside external staker capital in any loss event.

5.9 Net Supply Dynamics

The mechanisms above interact to produce a supply trajectory responsive to protocol activity and coverage state rather than a fixed curve:

- Scheduled emissions (5.2) set an upper bound on annual mint volume.
- Activity gating (5.5) withholds validator emissions when attestation volume is low; the withheld amounts are never minted.
- Below-minimum tranche emissions (5.7) are minted and converted to permanent coverage capital.
- Validator fee buyback (5.4) reduces supply in proportion to protocol economic activity.
- Waterfall mints (USDX whitepaper Section 8, Step 4) increase supply in loss events to recapitalize the reserve.

5.10 Summary

Supply dimension	Value
Initial supply	1,000,000,000 RATES
Year-one emission rate	12%
Annual emission rate decay	Multiplicative ratio of 0.88
Emissions split (validators / tranche stakers)	50% / 50%
Tranche sub-split (T1 / T2 / T3)	60% / 25% / 15%
Buyback-and-burn share of validator USDX fees	20%
T1 coverage band	0.5% – 10% of USDX supply
T2 coverage band	1% – 10% of USDX supply
T3 coverage band	2% – 10% of USDX supply

Asymptotic supply (emissions schedule alone)	2.0B (exactly)
--	----------------

6. Topics Deferred to Follow-Up Implementation Specification

The following operational topics are deferred to a follow-up implementation document published ahead of RATES distribution:

- RATES market-making and initial liquidity strategy
- Validator onboarding and KYC process
- Treasury management of the Stable Company and protocol-treasury allocations
- Cross-chain RATES representation and supply accounting

This document is a technical description of the RATES token as designed. It does not constitute an offer to sell, or the solicitation of an offer to buy, any security or financial instrument. RATES is a staking and governance token whose characterization is jurisdiction-specific.

Stable Company · Version 0.1 · June 2026 · trystable.co